

PRAGATI BHARATI - ADVANCED ALGORITHMS
QUESTION PAPER (Separate Answer Key Document Required)

1. What is the worst-case time complexity of QuickSort when using naive pivot selection?
(A) $O(n \log n)$
(B) $O(n^2)$
(C) $O(\log n)$
(D) $O(n)$
2. Which algorithm is best suited for finding the shortest path in a graph with non-negative edge weights?
(A) Bellman-Ford Algorithm
(B) Floyd-Warshall Algorithm
(C) Dijkstra's Algorithm
(D) Kruskal's Algorithm
3. Which dynamic programming algorithm solves the 0/1 Knapsack problem with weights W and N items?
(A) $O(N * W)$ pseudo-polynomial time
(B) $O(2^N)$ exponential time
(C) $O(N \log N)$ linearithmic time
(D) $O(W^2)$ polynomial time